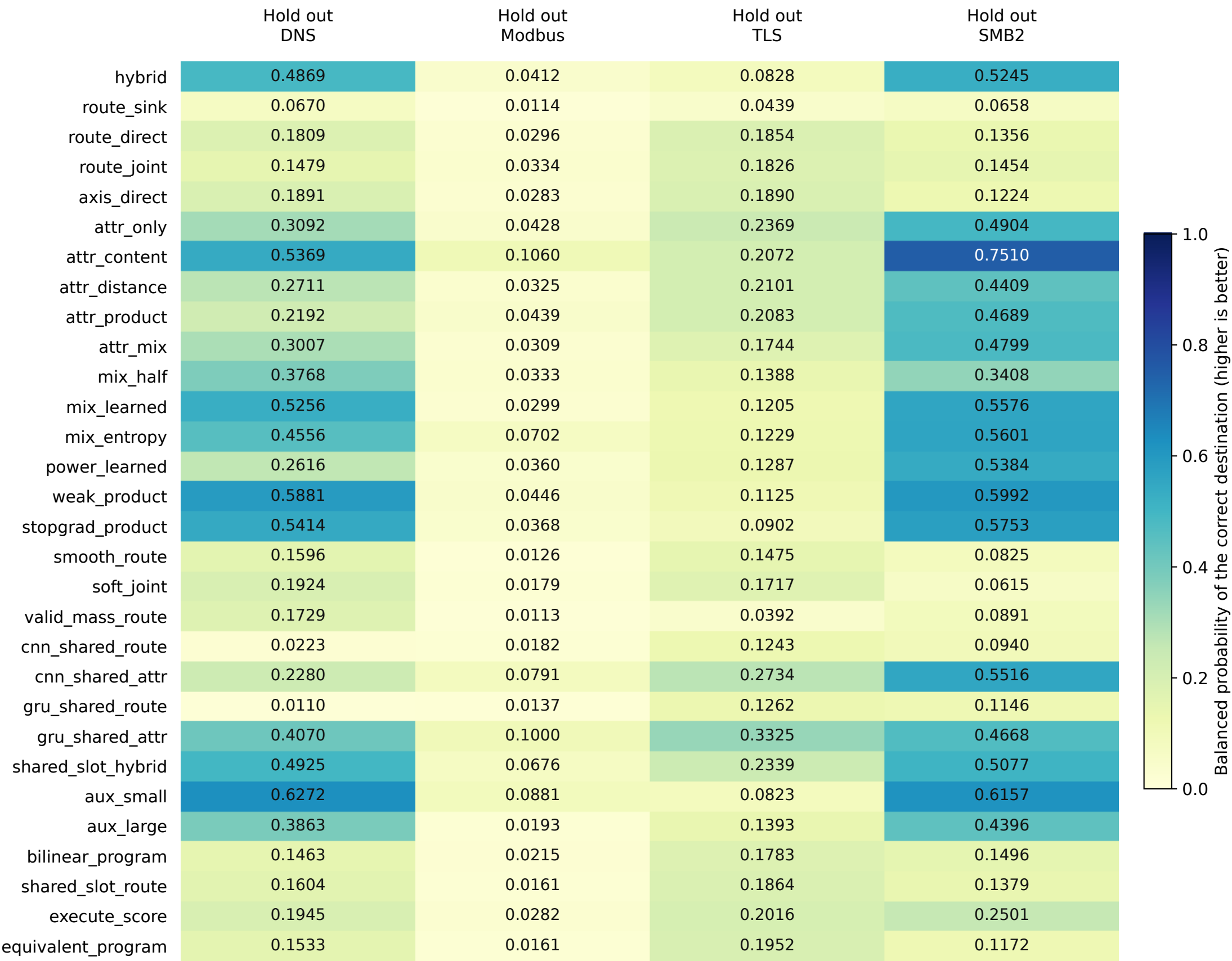


All 30 formulas: source-only screening

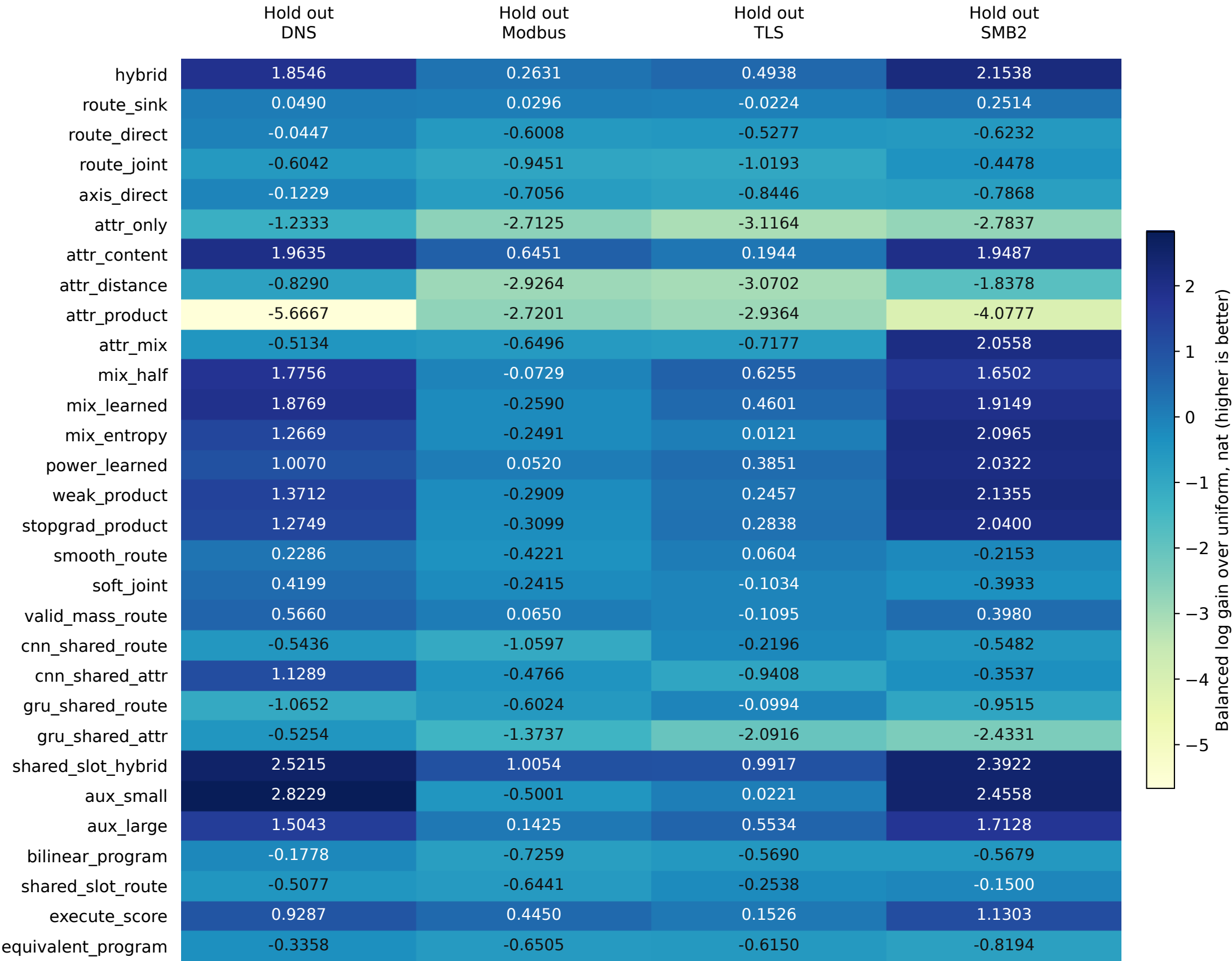
Balanced probability of the correct destination (higher is better)



120 updates; one seed; TAPE anchor (CNN/GRU: no Transformer backbone).
Each column uses only inner validation within its three source protocols; the held-out target is excluded.
These are selection-stage scores, not final test results.

All 30 formulas: source-only screening

Balanced log gain over uniform, nat (higher is better)



120 updates; one seed; TAPE anchor (CNN/GRU: no Transformer backbone).
Each column uses only inner validation within its three source protocols; the held-out target is excluded.
These are selection-stage scores, not final test results.

Source-only refinement and frozen formula selection

Held-out target	Refined formula	Balanced p	Log gain (nat)	Selected role
DNS	hybrid	0.3535	1.0926	-
DNS	aux_small	0.5336	2.3207	P + NLL
DNS	attr_content	0.4331	1.8667	-
DNS	shared_slot_hybrid	0.3476	0.7342	-
Modbus	hybrid	0.0298	-0.9131	NLL
Modbus	attr_content	0.0534	-2.1320	-
Modbus	shared_slot_hybrid	0.0522	-1.1839	-
Modbus	mix_entropy	0.0547	-1.5188	P
TLS	hybrid	0.0426	-0.5770	-
TLS	gru_shared_attr	0.2123	-3.1379	-
TLS	shared_slot_hybrid	0.0993	0.3003	NLL
TLS	attr_content	0.2447	-0.8814	P
SMB2	hybrid	0.3223	1.2964	-
SMB2	attr_content	0.5583	2.3979	P + NLL
SMB2	aux_small	0.3294	1.5435	-
SMB2	weak_product	0.5019	1.7623	-

400 updates from scratch; two fresh seeds; four promoted formulas per outer target.
P and NLL are selected independently. Selection is anchored to TAPE, not retuned per final backbone.

Final probability assigned to the correct destination

Mean p(target), higher is better



3 source protocols -> 1 held-out target; three seeds; 600 updates. Shown: raw field means, not balanced selection scores.
P-selected / NLL-selected formulas vary by target (see selection table); no target-based reselection.
Any CNN/GRU champion is one canonical model reused across backbone panels. Tiny nonzero values use scientific notation.

Final destination retrieval: Hit@1

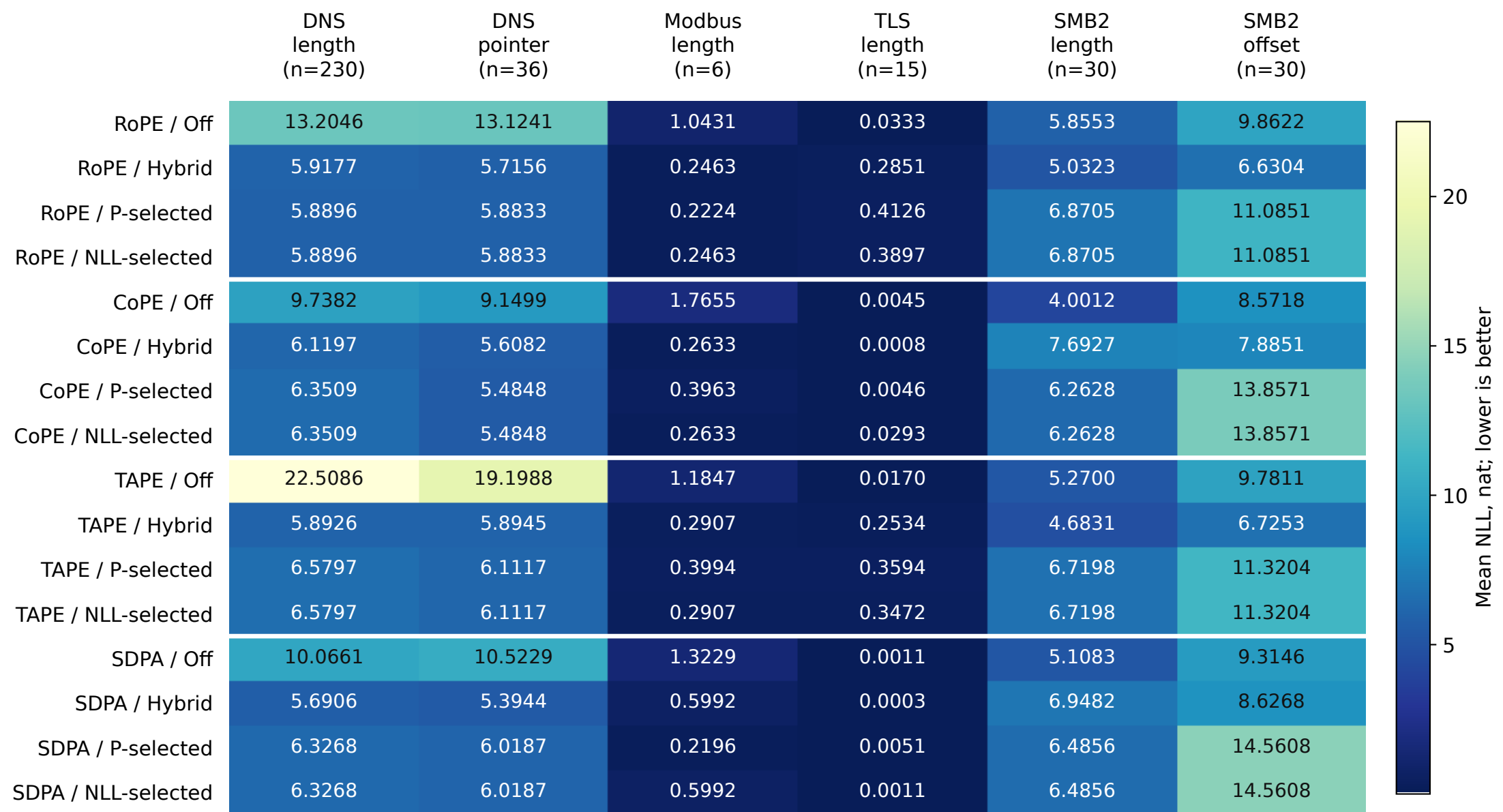
Mean Hit@1, higher is better



3 source protocols -> 1 held-out target; three seeds; 600 updates. Shown: raw field means, not balanced selection scores.
P-selected / NLL-selected formulas vary by target (see selection table); no target-based reselection.
Any CNN/GRU champion is one canonical model reused across backbone panels. Tiny nonzero values use scientific notation.
Uniform Hit@1 uses deterministic argmax at byte 0, not sampled-hit expectation.

Final destination negative log-likelihood

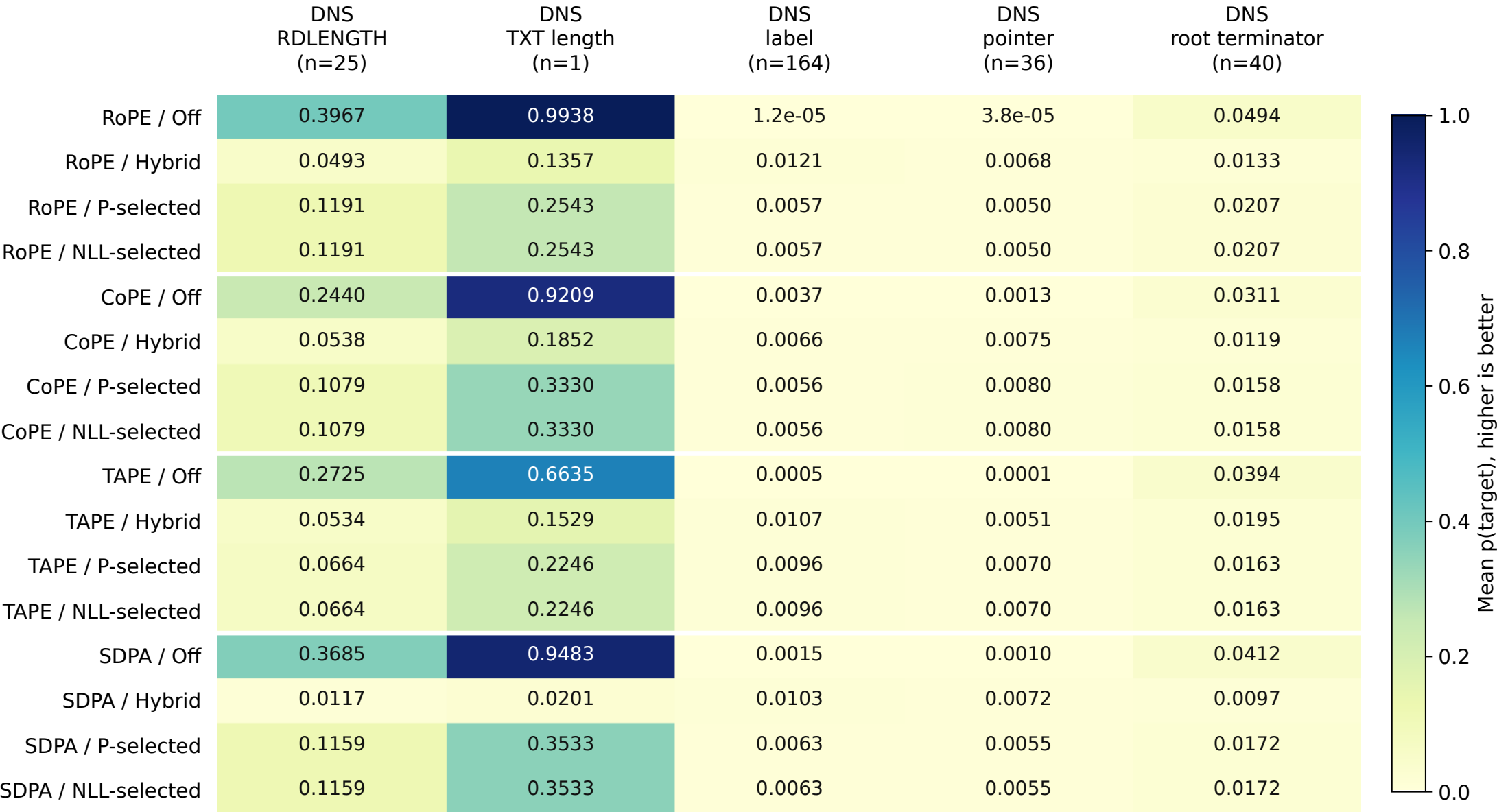
Mean NLL, nat; lower is better



3 source protocols -> 1 held-out target; three seeds; 600 updates. Shown: raw field means, not balanced selection scores.
P-selected / NLL-selected formulas vary by target (see selection table); no target-based reselection.
Any CNN/GRU champion is one canonical model reused across backbone panels. Tiny nonzero values use scientific notation.

DNS: separate address relations

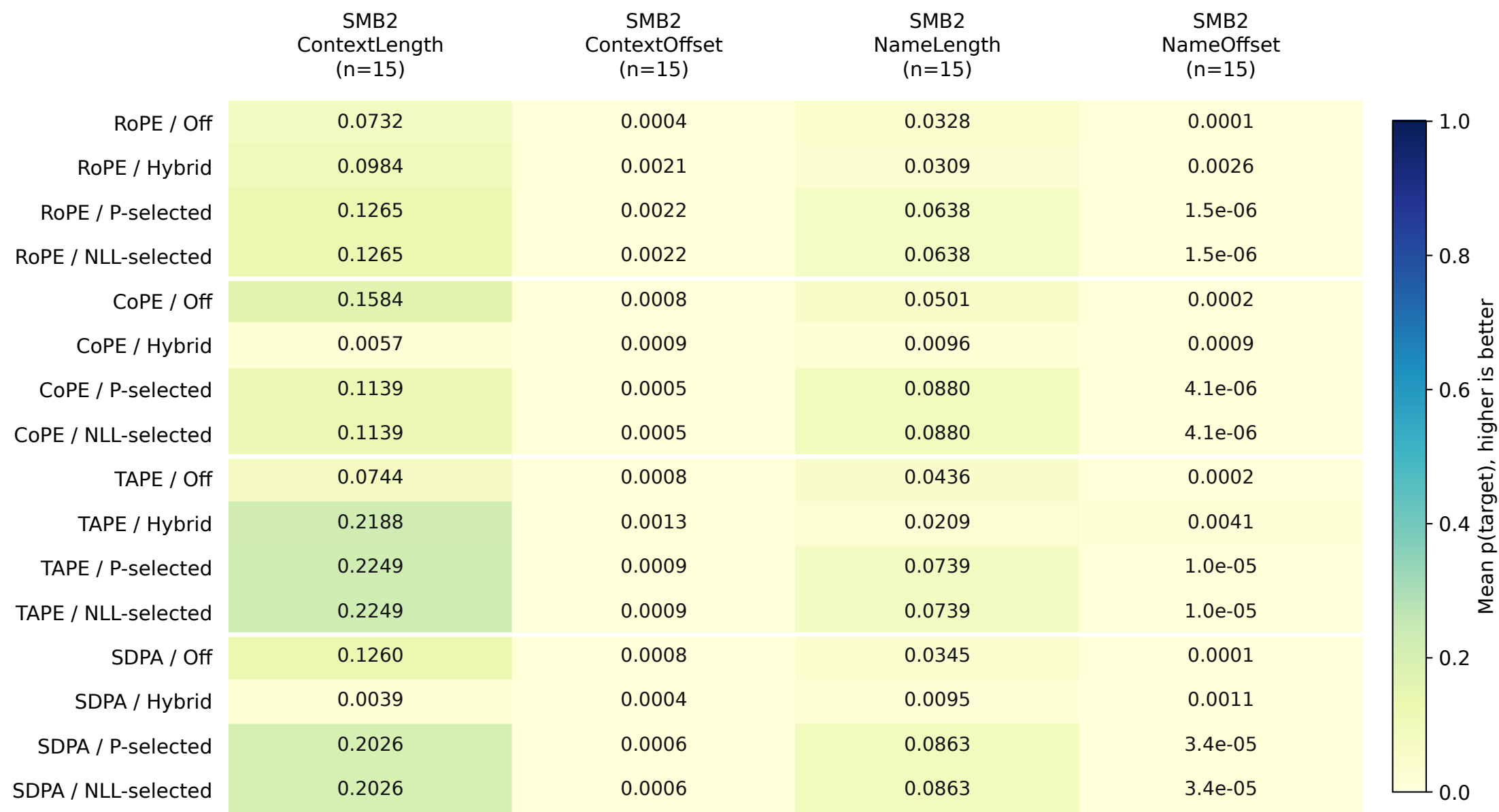
Mean p(target), higher is better



3 source protocols -> 1 held-out target; three seeds; 600 updates. Shown: raw field means, not balanced selection scores.
P-selected / NLL-selected formulas vary by target (see selection table); no target-based reselection.
Any CNN/GRU champion is one canonical model reused across backbone panels. Tiny nonzero values use scientific notation.

SMB2: separate address relations

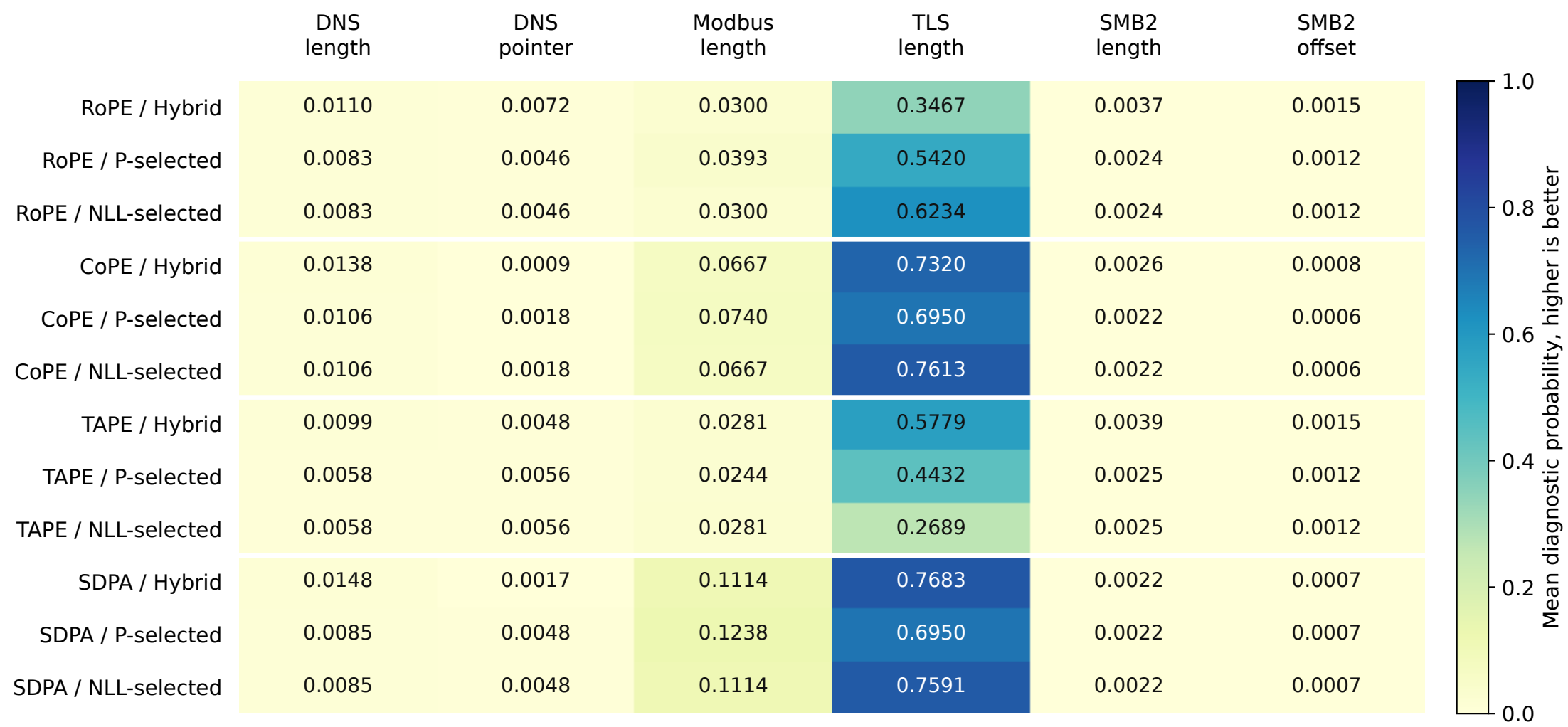
Mean p(target), higher is better



3 source protocols -> 1 held-out target; three seeds; 600 updates. Shown: raw field means, not balanced selection scores.
P-selected / NLL-selected formulas vary by target (see selection table); no target-based reselection.
Any CNN/GRU champion is one canonical model reused across backbone panels. Tiny nonzero values use scientific notation.

Source localization probability

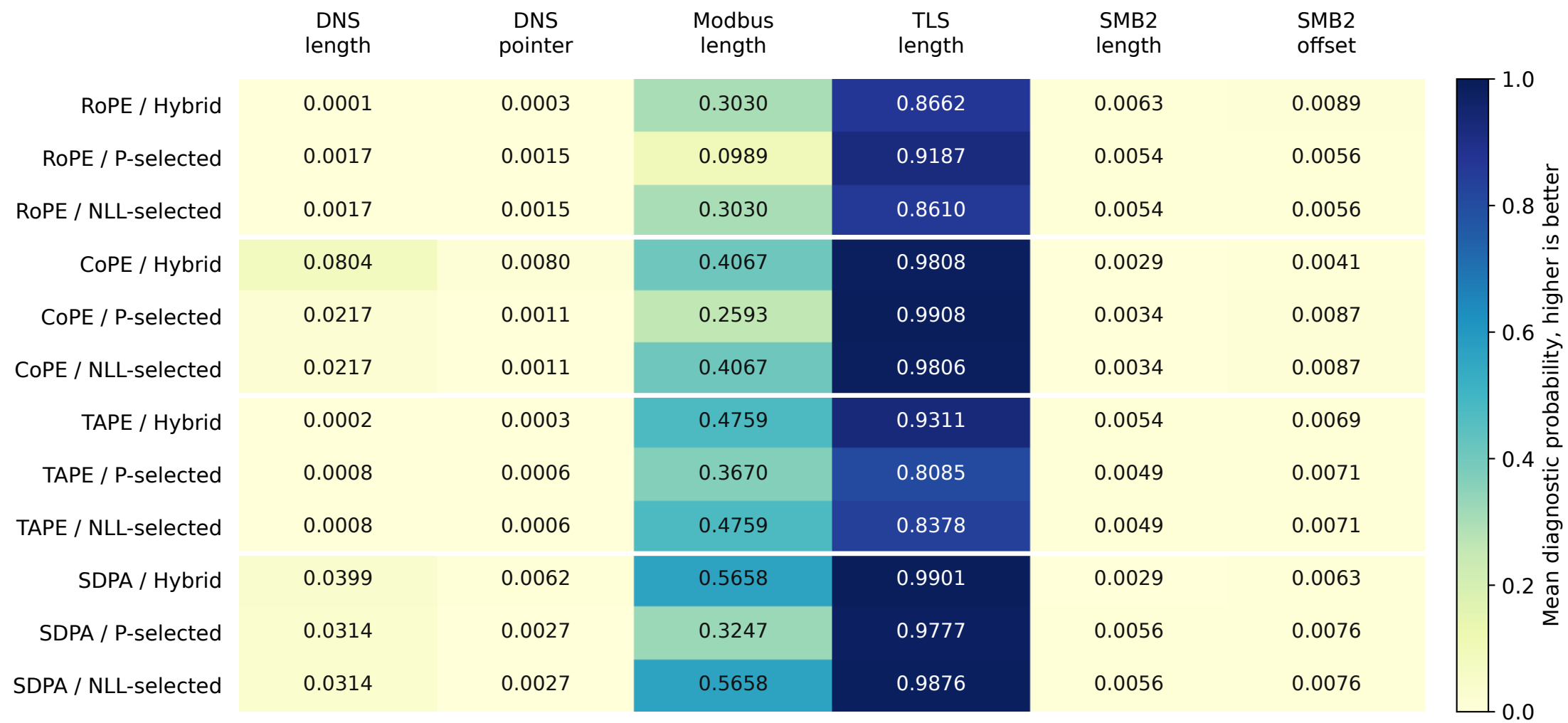
Mean diagnostic probability, higher is better



Gold source is used only for post-seal diagnostics, never as a model input.
Program correctness uses the labeled program; alternative programs can have the same executed destination.
Pre-validity head probabilities, not normalized route posteriors. Off heads are untrained and omitted; three-seed means.

Program probability conditioned on the true source

Mean diagnostic probability, higher is better



Gold source is used only for post-seal diagnostics, never as a model input.
Program correctness uses the labeled program; alternative programs can have the same executed destination.
Pre-validity head probabilities, not normalized route posteriors. Off heads are untrained and omitted; three-seed means.

Exact typed-field discovery: F1

Mean exact typed-span F1, higher is better

	DNS length	DNS pointer	Modbus length	TLS length	SMB2 length	SMB2 offset
RoPE / Off	N/A	N/A	N/A	N/A	N/A	N/A
RoPE / Hybrid	0.0000	0.0000	0.0000	0.3794	0.0000	0.0000
RoPE / P-selected	0.0000	0.0000	0.0000	0.6716	0.0000	0.0000
RoPE / NLL-selected	0.0000	0.0000	0.0000	0.4557	0.0000	0.0000
CoPE / Off	N/A	N/A	N/A	N/A	N/A	N/A
CoPE / Hybrid	0.0000	0.0000	0.0000	0.8576	0.0000	0.0000
CoPE / P-selected	0.0000	0.0000	0.0000	0.8708	0.0000	0.0000
CoPE / NLL-selected	0.0000	0.0000	0.0000	0.6831	0.0000	0.0000
TAPE / Off	N/A	N/A	N/A	N/A	N/A	N/A
TAPE / Hybrid	0.0000	0.0000	0.0000	0.6594	0.0000	0.0000
TAPE / P-selected	0.0000	0.0000	0.0000	0.5879	0.0000	0.0000
TAPE / NLL-selected	0.0000	0.0000	0.0000	0.2892	0.0000	0.0000
SDPA / Off	N/A	N/A	N/A	N/A	N/A	N/A
SDPA / Hybrid	0.0000	0.0000	0.0000	0.9201	0.0000	0.0000
SDPA / P-selected	0.0000	0.0000	0.0000	0.8722	0.0000	0.0000
SDPA / NLL-selected	0.0000	0.0000	0.0000	0.8006	0.0000	0.0000

1.0

0.8

0.6

0.4

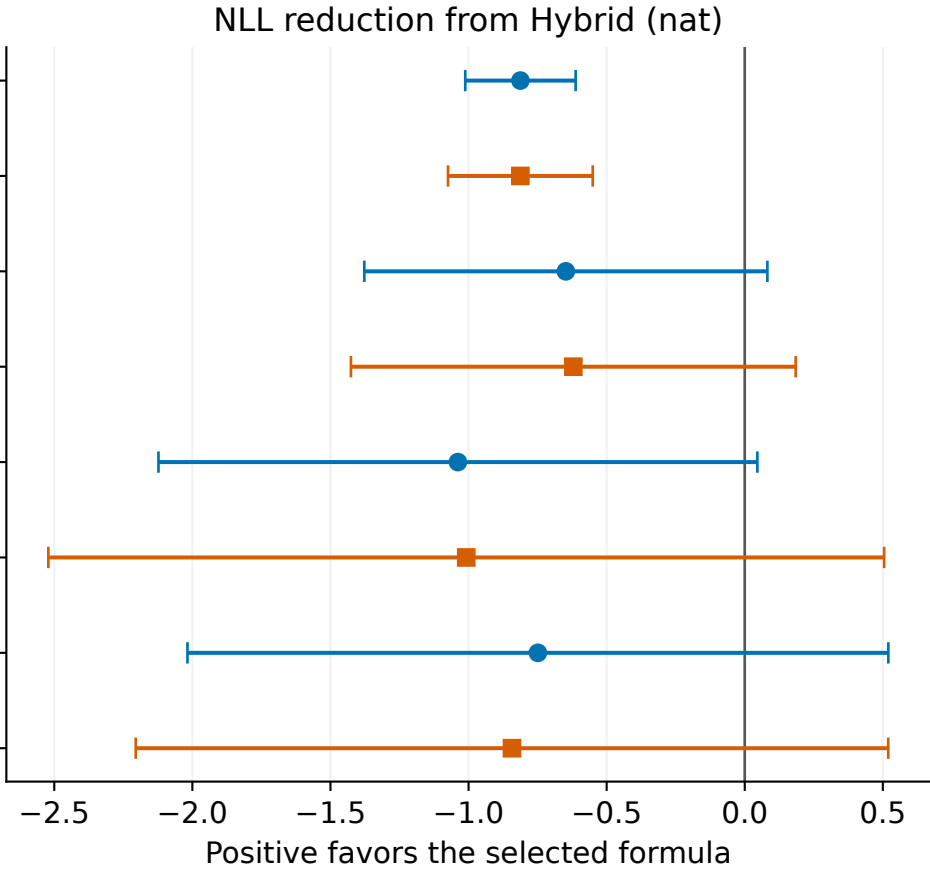
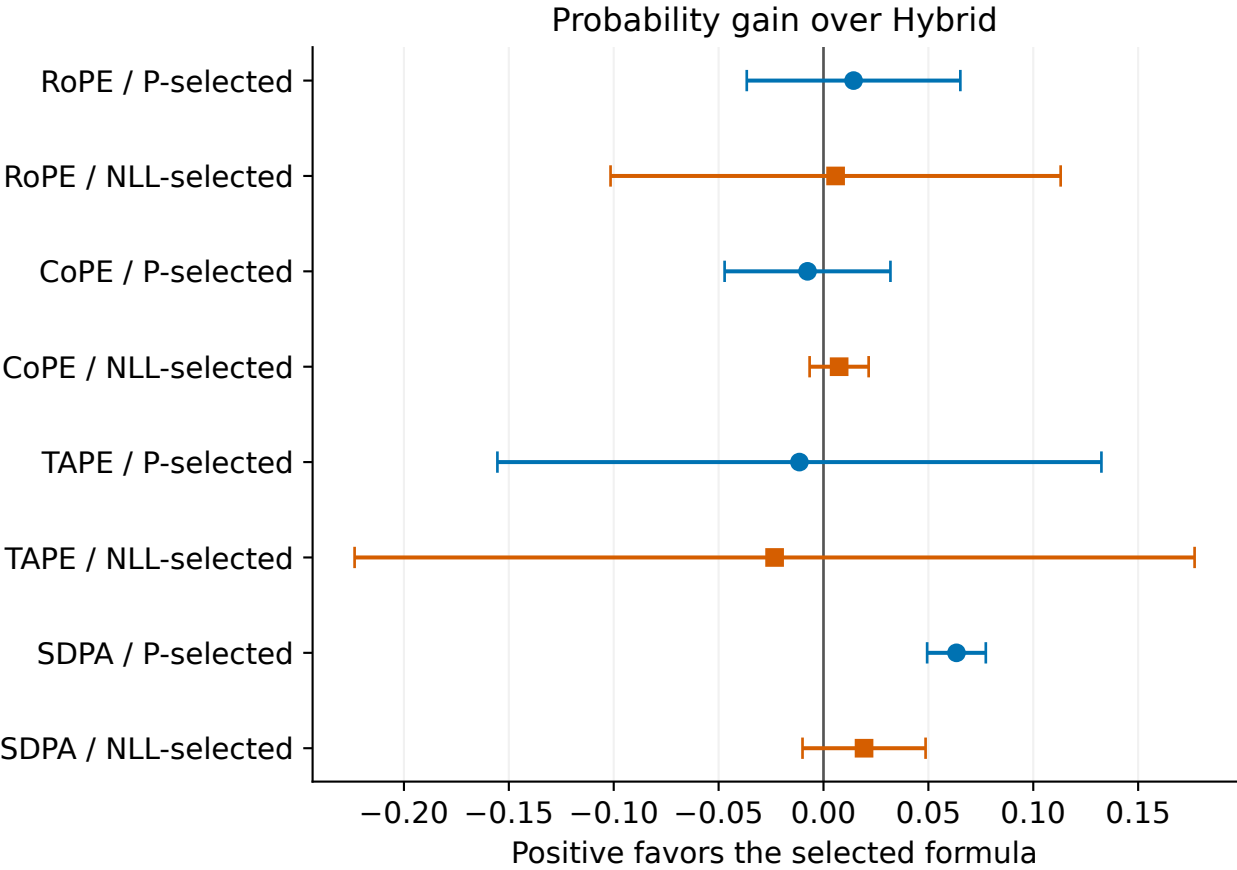
0.2

0.0

Mean exact typed-span F1, higher is better

Threshold chosen on source development data only; three-seed means.
Off has endpoint NLL only: its untrained field head is N/A, not a measured zero.
A destination probability is not a field-discovery F1; these metrics answer different questions.

Paired changes: equal-protocol macro average



Points: paired three-seed means. Whiskers: un-clipped 95% Student-t intervals (df=2).
Each protocol receives equal weight; fields are averaged within protocol.
Intervals describe training-seed variability, not uncertainty over captures or unseen protocols.
Blue circles: P-selected; orange squares: NLL-selected. Selection is source-only and TAPE-anchored.